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LAB NO 05

Operating System Lab

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Class: BSCS F'24 B

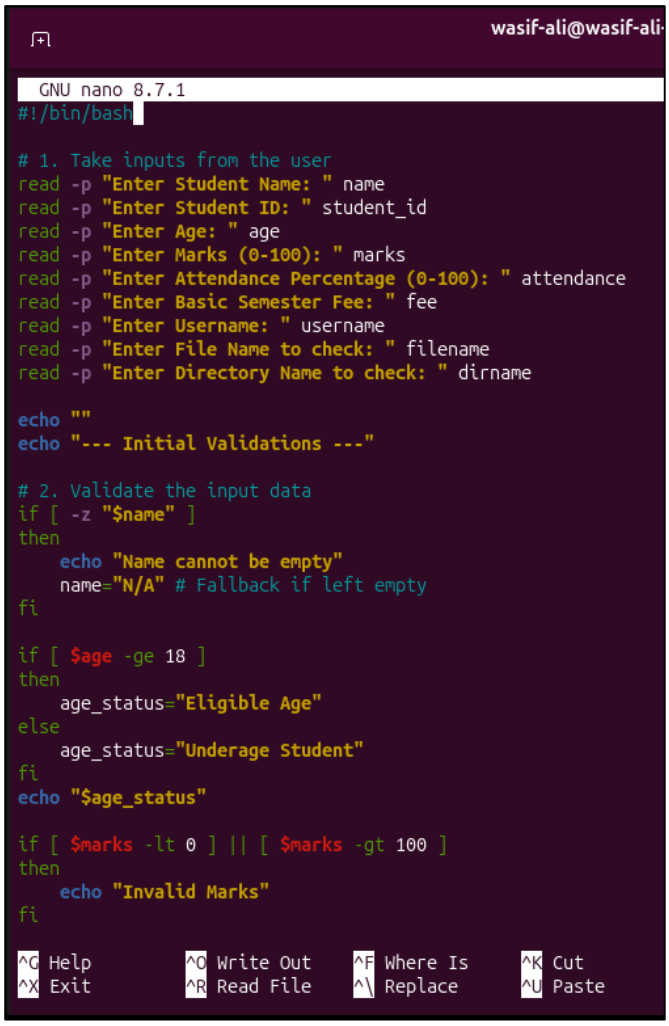
LAB TASK

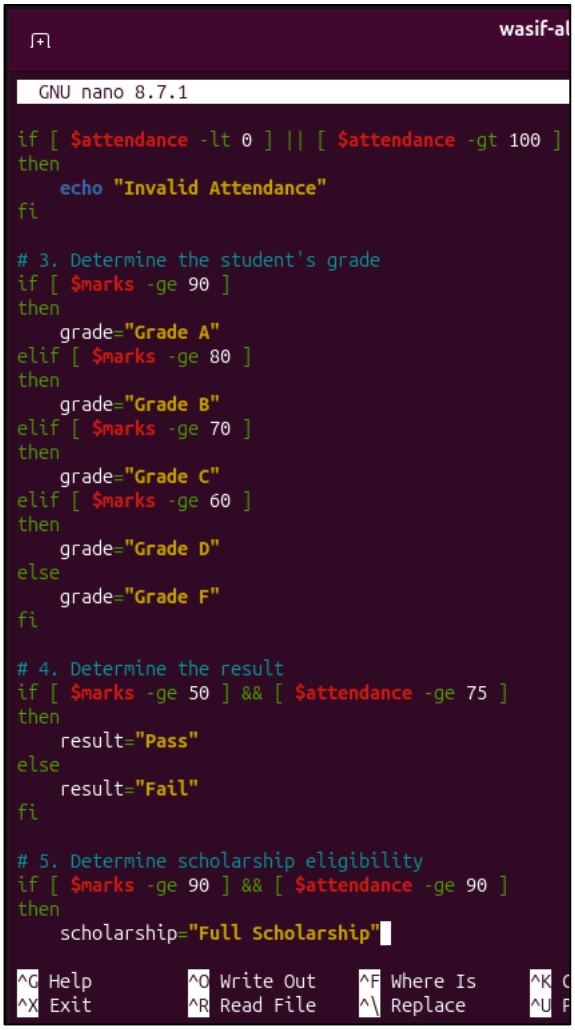
Lab Task — University Student Portal & Fee Management System
Problem Statement

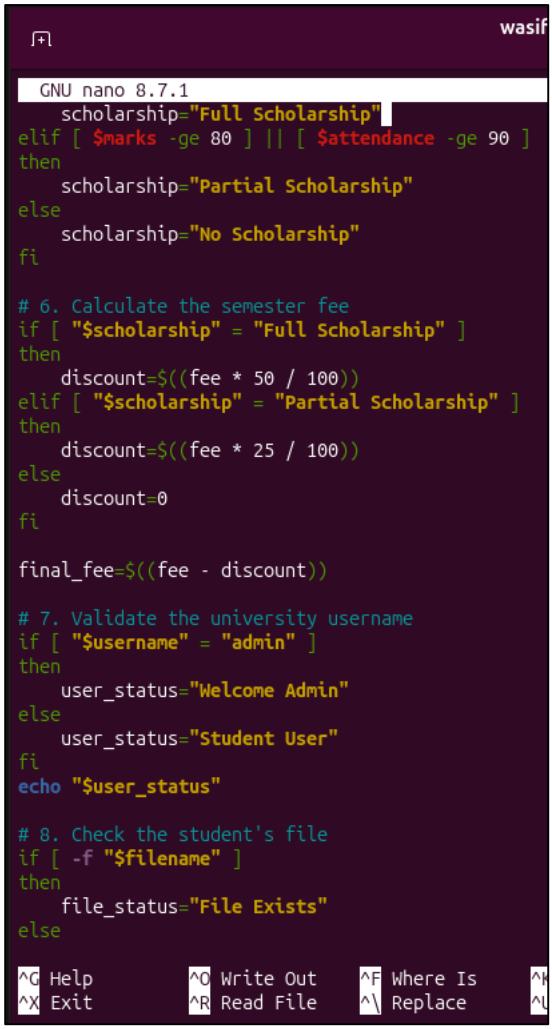
Prepare a menu-free shell script that simulates a basic **University Student Portal**. The script should take student information and perform different validations, calculate the student's result and fee status, and generate a final report.
The script should:

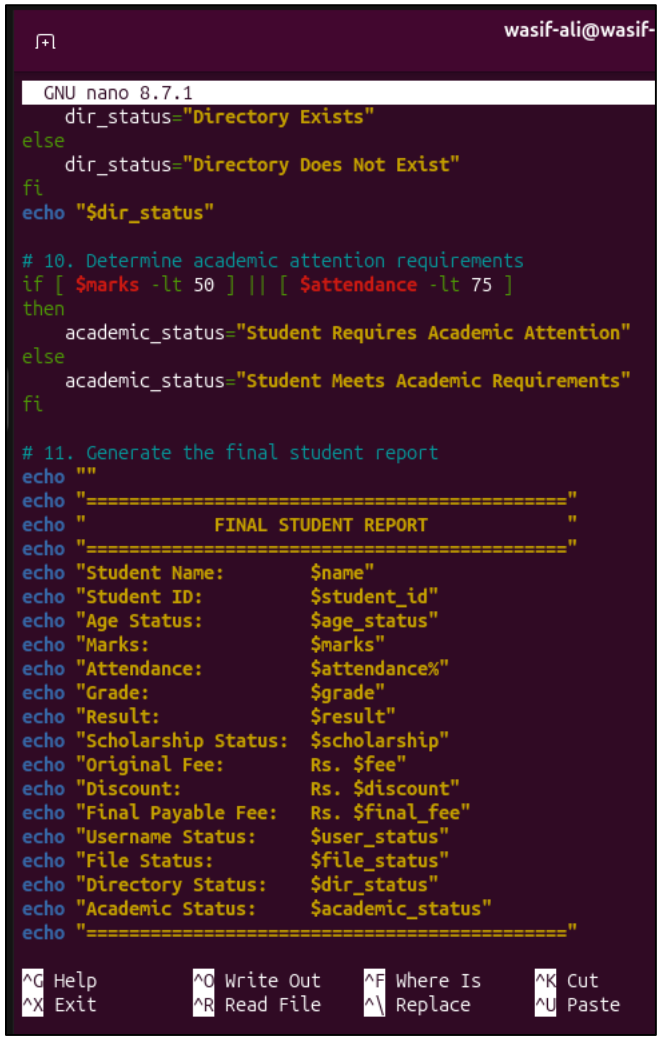
Requirements:
The solution must **demonstrate** the use of if, if-else, if-elif-else, numeric comparison operators (-eq, -ne, -gt, -lt, -ge, -le), string comparison, -z, -f, -d, logical operators && and ||, and arithmetic expressions in a single shell script.

Script File (Shell Script Code)**:**

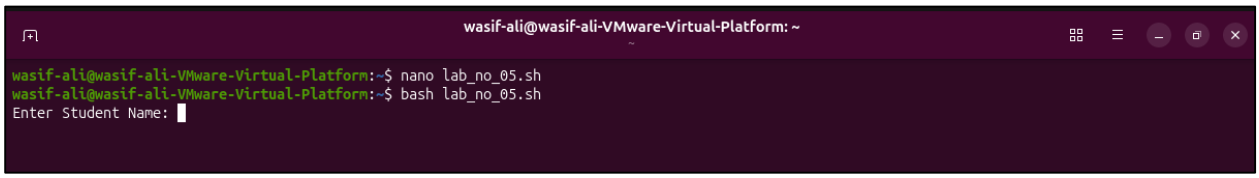


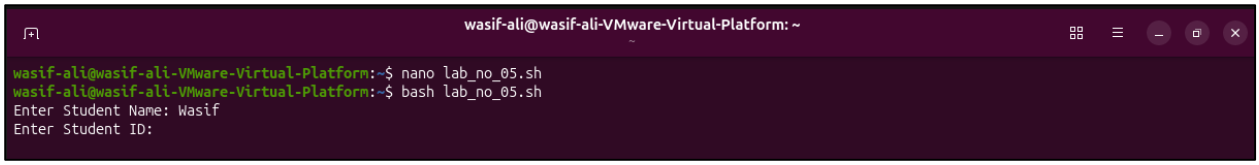


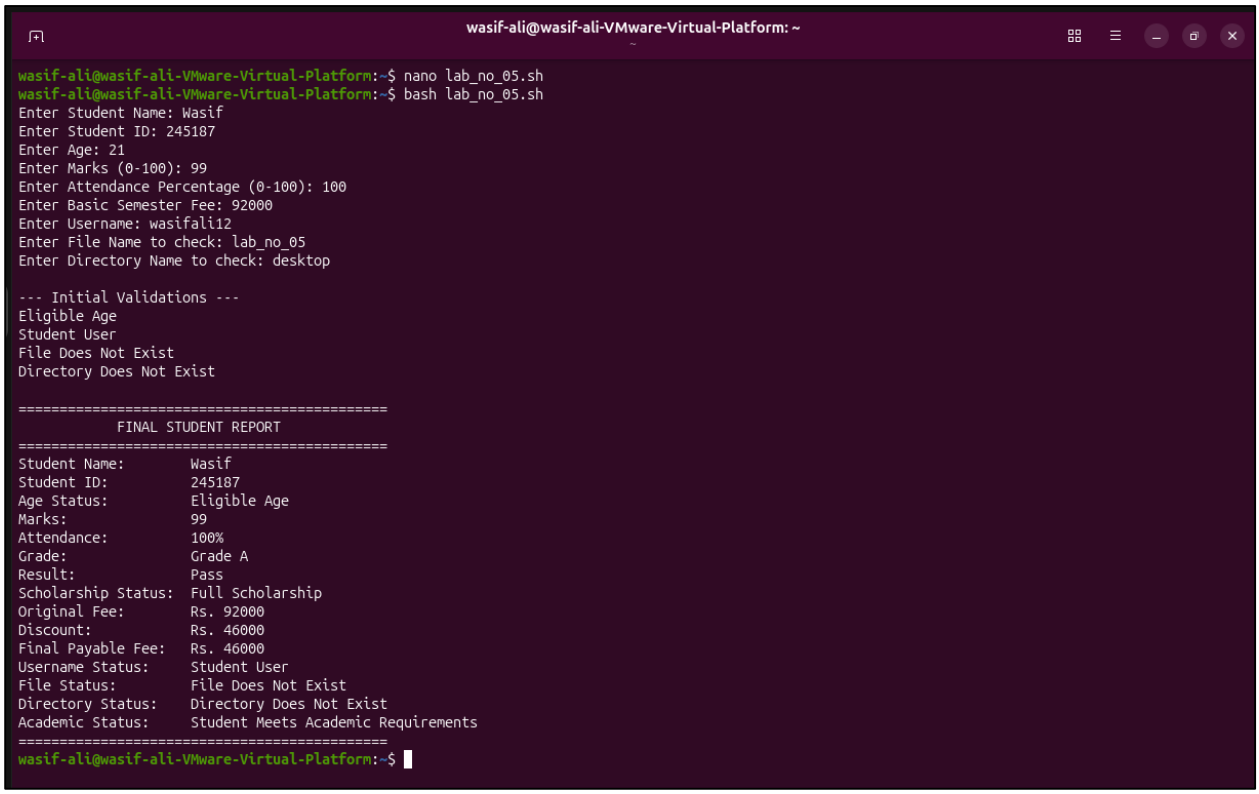

wasif
GNU nano 8.7.1
scholarship="Full Scholarship"
elif [$marks -ge 80] || [$attendance -ge 90]
then
scholarship="Partial Scholarship"
else
scholarship="No Scholarship"
fi
6. Calculate the semester fee
if ["$scholarship" = "Full Scholarship"]
then
discount=$((fee * 50 / 100))
elif ["$scholarship" = "Partial Scholarship"]
then
discount=$((fee * 25 / 100))
else
discount=0
fi
final_fee=$((fee - discount))
7. Validate the university username
if ["$username" = "admin"]
then
user_status="Welcome Admin"
else
user_status="Student User"
fi
echo "$user_status"
8. Check the student's file
if [-f "$filename"]
then
file_status="File Exists"
else
^G Help ^O Write Out ^F Where Is
^X Exit ^R Read File ^\ Replace


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GNU nano 8.7.1
dir_status="Directory Exists"
else
dir_status="Directory Does Not Exist"
fi
echo "$dir_status"
10. Determine academic attention requirements
if [$marks -lt 50] || [$attendance -lt 75]
then
academic_status="Student Requires Academic Attention"
else
academic_status="Student Meets Academic Requirements"
fi
11. Generate the final student report
echo ""
echo "==="
echo "   FINAL STUDENT REPORT   "
echo "==="
echo "Student Name: $name"
echo "Student ID: $student_id"
echo "Age Status: $age_status"
echo "Marks: $marks"
echo "Attendance: $attendance%"
echo "Grade: $grade"
echo "Result: $result"
echo "Scholarship Status: $scholarship"
echo "Original Fee: Rs. $fee"
echo "Discount: Rs. $discount"
echo "Final Payable Fee: Rs. $final_fee"
echo "Username Status: $user_status"
echo "File Status: $file_status"
echo "Directory Status: $dir_status"
echo "Academic Status: $academic_status"
echo "==="
^G Help ^O Write Out ^F Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste

Command







University Student Portal & Fee Management System Description

- **Data Collection:** Prompts the user to enter nine specific data points, including academic performance, financial details, and system file names.
- **Input Validation:** Verifies that the student's name is not left empty using the `-z` flag, categorizes the student's age (18+ as eligible), and identifies if the entered marks or attendance fall outside the valid 0-100 range.
- **Academic Grading & Results:** Uses an `if-elif-else` ladder to assign a letter grade (A through F) based on marks, and uses the logical `&&` operator to grant a "Pass" status only if marks are at least 50 and attendance is at least 75%.
- **Scholarship & Fee Calculation:** Evaluates scholarship tiers using both `&&` (Full Scholarship for 90+ marks and 90%+ attendance) and `||` (Partial Scholarship for 80+

marks or 90%+ attendance) operators, subsequently calculating a 50% or 25% fee discount using arithmetic expressions.

- **System Verification:** Compares the entered username to "admin" to determine access level, and utilizes the `-f` and `-d` flags to confirm whether the specified file and directory actually exist on the system.
- **Final Reporting:** Flags the student for academic attention if they fail to meet minimum marks or attendance thresholds, and prints a consolidated, formatted summary report of all evaluated data.